

Customer Churn Data Analytics Project

Stage 3 Summary: Data Understanding

This document summarizes the Data Understanding stage of the Customer Churn Data Analytics Project, based on the diagnostic notebook *03. Data Understanding*. The purpose of this stage is to discover and document what is actually inside the dataset before any cleaning or transformation takes place. No values are modified, no problems are fixed, and no business KPIs are formally interpreted at this point — only observed and recorded.

This stage is diagnostic only. The raw dataset (`raw_df`) remains untouched throughout the notebook. Any anomalies identified here are documented for resolution in Stage 4 — Data Quality Assessment, not corrected now.

1. Stage Activities

Step	Activity	Description
3.1	Load the Raw Dataset	Load the untouched dataset into a DataFrame.
3.2	Inspect Dataset Dimensions	Determine number of rows and columns.
3.3	Inspect Column Names	Verify all variables and detect unexpected naming issues.
3.4	Inspect Sample Records	Examine the first and last records to understand data representation.
3.5	Inspect Data Types	Identify numerical, categorical, boolean/flag, and object/string columns.
3.6	Understand Each Variable's Values	Examine unique values and category distributions for categorical variables.
3.7	Understand Numerical Variables	Inspect ranges, central tendency, and spread of numerical variables.
3.8	Understand Target Variable	Examine the distribution of churn / churn_flag.
3.9	Identify Potential Anomalies	Look for suspicious values, unexpected categories, or inconsistent representations, without correcting them.
3.10	Understand Dataset Granularity	Confirm that each row represents one customer and customerid identifies the customer.
3.11	Compare With Data Dictionary	Verify that the actual dataset matches the documentation created in Stage 2.
3.12	Document Findings	Record what was learned about structure, variables, distributions, and potential issues.

2. Diagnostic Workflow (Notebook Overview)

The accompanying notebook loads the raw CSV file (`WA_Fn-UseC_-Telco-Customer-Churn.csv`) into a reference DataFrame, `raw_df`, which is kept unmodified throughout the project. The workflow proceeds through the following diagnostic checks:

Structure & Sample Inspection

- Dataset dimensions: number of rows (customers) and columns (variables), via `df.shape`.
- Column names and data types, non-null counts, and memory usage, via `df.info()`.
- First and last records, via `df.head()` and `df.tail()`, to confirm the one-row-per-customer structure and see how values are represented.
- Cardinality of every field, via `df.nunique()`, to get a quick view of which columns are truly categorical.

Categorical Variables

Key categorical dimensions used in the business questions — `Contract`, `InternetService`, `PaymentMethod`, and `Churn` — are explicitly inspected for their actual unique values (rather than assuming the documented categories match the file exactly), and their frequency distributions are reviewed with `value_counts()`.

Target Variable (Churn)

The `Churn` column is inspected for both raw counts and normalized percentages, establishing the baseline class distribution (churned vs. retained) as a diagnostic observation. This is distinct from the formal churn-rate KPI calculation and business interpretation, which belong to a later analysis stage.

Cross-Tabulations (Preliminary)

Preliminary cross-tabulations of `Churn` against `Contract`, `InternetService`, and `PaymentMethod` (using `normalize="index"`) are produced to see how churn is distributed within each category. These are early descriptive checks, not final conclusions.

Numerical Variables

`Tenure`, `MonthlyCharges`, and `TotalCharges` are summarized with descriptive statistics (minimum, maximum, mean, median, standard deviation, quartiles) via `df.describe()`, and their distributions are visualized with histograms. Boxplots are also produced for all three variables to help identify unusually high or low observations.

3. Potential Anomalies Identified (Not Yet Corrected)

The following observations were noted during diagnostic inspection. Per the Data Understanding stage guidelines, these are documented only — no corrections are applied here.

- `TotalCharges`: in the standard Telco churn dataset, this field is commonly stored as an object/string data type rather than numeric, typically due to blank string values for customers with very low tenure. A temporary numeric conversion (via `pd.to_numeric` with `errors="coerce"`) is used only for inspection purposes, without altering the raw column.
- Outliers in `tenure`, `MonthlyCharges`, or `TotalCharges` observed in the boxplots are not assumed to be errors — they are flagged as candidates for review in the Data Quality Assessment stage.
- Any unexpected categories or inconsistent spellings surfaced in `unique()` outputs for categorical fields (e.g., `Contract`, `InternetService`, `PaymentMethod`) should be recorded here rather than corrected.

Example of correct framing at this stage: *"TotalCharges is stored as an object/string data type and may require conversion."* The conversion itself is deferred to Stage 4 — Data Quality Assessment.

4. Dataset Granularity

Inspection of sample records confirms that each row represents one customer, and customerid serves as the unique identifier for that customer. This is consistent with the granularity documented in Stage 2.

5. Comparison With Data Dictionary

Column names, data types, and category values observed in the raw dataset should be cross-checked against the data dictionary produced in Stage 2 (Data Acquisition & Documentation) to confirm that the actual file matches what was documented, and to flag any discrepancies for follow-up.

6. Deliverable

A written record of findings from this stage — covering dataset structure, variable types, category and numerical distributions, target distribution, granularity confirmation, and any potential anomalies — to be documented in Word/Google Docs and carried forward into the Data Quality Assessment stage.

Prepared as part of the Customer Churn Data Analytics Project — Data Understanding stage.